

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester IV Examination, 2025-26 (NEP)

BOTANY**Paper No. BOTMJ43 : Gymnosperms and Paleobotany**

Ref. No.: Conf/Even/N/2025-26/20

Time: 2½ hours

Full Marks: 70

*(Write your Roll No. at the top immediately on the receipt of this question paper)**The figures in the right-hand margin indicate marks.**Answer **five** questions including Question 1 which is compulsory.***Question 1.** Answer the following questions in not more than 100 words each: [$7 \times 2 = 14$]

- (a) Why are the leaves of *Cycas* described as pinnately compound but fern-like in appearance? [2]
- (b) Describe the salient features of Progymnosperm. [2]
- (c) Why are the leaves of *Ephedra* reduced and what is the role of its green, jointed stems? [2]
- (d) Give an account of the affinities of Pteridospermales with Cycadophyta and ferns. [2]
- (e) Why are the seeds of *Pinus* winged and how are these wings derived? [2]
- (f) Describe the anatomical features of *Lyginopteris*. [2]
- (g) What is the significance of the presence of resin canals for survival in *Pinus*? [2]

Question 2. Describe the geological time scale, outlining its major divisions, key events in each era and its significance in understanding the evolution of plant life. [14]**Question 3.** Write short notes on **any two** of the following: [$2 \times 7 = 14$]

- (a) Reproductive parts of *Lyginopteris* [7]
- (b) Pteridospermales [7]
- (c) Classification of gymnosperms [7]

Question 4. Give a comprehensive account of the types of fossils and their importance, with suitable diagrams. [14]

Question 5. Write short notes on **any two** of the following: [2 × 7 = 14]

- (a) Reproduction in *Pinus* [7]
 - (b) Difference between young and old stems of *Ephedra* [7]
 - (c) Structure and adaptations of *Pinus* leaf [7]
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Question 6. Describe the anatomy and reproduction of *Cycas*, highlighting its distinctive features and evolutionary significance. [14]

Question 7. Write short notes on **any two** of the following: [2 × 7 = 14]

- (a) Major geological events in Earth's history [7]
- (b) *Glossopteris* leaf [7]
- (c) Poikilohydry and homoiohydricity [7]